

The finite option case as an unconstrained maximization problem

Consider a financial market defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P}) = (\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2), \bar{\mu})$ with d assets with **random payoffs** $(S^1, \dots, S^d)$ of the form $g_i(X, Y)$ at time T with $S^0 = 1$ (i.e. cash) and $S^1 = X$ and $S^2 = Y$ (i.e. forwards on X and Y resp.) and assume all payoffs are **linearly independent** with **market prices** π_i at $t = 0$ (these assets can include **European call/put options**). For notational convenience, we let $\tilde{Y}_i = S_i - \pi_i$, and assume a financial agent can only trade at time zero.

Derive the **first order optimality condition** for an agent to maximize their **expected utility** $\mathbb{E}(U(b \cdot \tilde{Y}))$ over $b \in \mathbb{R}^d$, where $U : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing function with $U''(x) < 0$ and b_i is the position in the i 'th asset and we assume the **moment condition** $\mathbb{E}(U(b \cdot \tilde{Y})) < \infty$ for all $b \in \mathbb{R}^d$.

Solution. As in first year calculus, we compute derivatives wrt each b_i and then set the answer to zero:

$$\frac{\partial}{\partial b_i} \mathbb{E}(U(b \cdot \tilde{Y})) = \mathbb{E}(\tilde{Y}_i U'(b \cdot \tilde{Y})) = 0$$

for $i = 1..d$, i.e. we have d equations for the d unknowns $b_1^*, \dots, b_d^*$ for the optimal portfolio allocation b^* . Note we can re-write this as

$$\mathbb{E}^{\mathbb{Q}}(\tilde{Y}_i) = 0 \tag{1}$$

where we define a **new probability measure** as $\mathbb{Q}(A) := \mathbb{E}^{\mathbb{P}}\left(\frac{U'(b^* \cdot \tilde{Y})}{\mathbb{E}^{\mathbb{P}}(U'(b^* \cdot \tilde{Y}))} 1_A\right)$ for events $A \in \mathcal{F}$, and (as a sanity check) we note that $\mathbb{Q}(\Omega) = 1$.

Under the moment condition stated in the question, it turns out that a unique solution b^* exists if the **no-arbitrage** condition is satisfied: if $\mathbb{P}(b \cdot \tilde{Y} > 0) > 0$ then $\mathbb{P}(b \cdot \tilde{Y} < 0) > 0$. In this case, from (1), we see that under $\mathbb{Q}$, all contracts are priced according to the market, i.e. $\mathbb{E}^{\mathbb{Q}}(S_i) = \pi_i$. $\mathbb{Q}$ is known as a **risk-neutral measure**. We can solve this maximization problem numerically using e.g. MOSEK convex optimization package in Python (used for summer project).

A common choice is the **exponential utility function** $U(x) = -e^{-\lambda x}$ so $U'(x) = \lambda e^{-\lambda x}$ and W.L.O.G we can set $\lambda = 1$, so in this case

$$\mathbb{Q}(A) = \mathbb{E}^{\mathbb{P}}\left(\frac{e^{-b^* \cdot \tilde{Y}}}{\mathbb{E}^{\mathbb{P}}(e^{-b^* \cdot \tilde{Y}})} 1_A\right) = \int_A \frac{e^{-\sum_i b_i g_i(x, y)}}{\int \int e^{-\sum_i b_i g_i(x, y)} dxdy} \bar{\mu}(x, y) dxdy \tag{2}$$

so the optimal probability density can be read off as

$$\mu^*(x, y) = \frac{e^{-\sum_i b_i g_i(x, y)}}{(...)} \bar{\mu}(x, y)$$

where (...) denotes the double integral in (2), which is needed to “normalize” μ^* to be a probability density.

Note we are computing $\max_b(-\mathbb{E}(e^{-b \cdot \tilde{Y}})) = \max_b(-\mathbb{E}(e^{b \cdot \tilde{Y}})) = -\min_b \mathbb{E}(e^{b \cdot \tilde{Y}})$, i.e. minus the **minimum of the mgf** of $\tilde{Y}$.

The maximization here can be easily done in Python or (quicker) using MOSEK in Python (as used for Project 2), which requires obtaining a licence.